



Fig. 6: Pattern to unlock an Android phone



Fig. 7: SecurID e-tokens (connected and disconnected types) (Credit: RSA SecurID)

nism for authentication. Many TFA techniques rely on password as one factor of authentication.

PIN. A PIN is a secret numeric password, typically used in ATMs.

Pattern. A pattern is a sequence of cells in an array used for authenticating users, such as pattern-based authentication used in Android devices (Fig. 6).

What you have

A user has to use what he or she has for authentication, such as an ATM card, mobile phone, e-token, etc. During authentication, websites or services will send a temporary password, OTP, to your registered mobile phone number. Some e-token devices (like connected tokens, where they are required to connect with the system) are just like pen drive dongles used to connect with the system. An e-token uses a digital signature or some kind of a secret key during authentication only. It is very difficult to use those keys outside the application. Other e-token devices (like disconnected tokens in Fig. 8, where they are not required to

Authentication Methods Listed with Examples and Their Properties

Authentication method	Examples	Properties
What you know (something user knows)	User IDs	Shared
	PINs	Easy to guess or predict
	Passwords	Usually forgotten as complexity grows
What you have (something user owns)	Cards	Shared
	Badges	Can be duplicated
	Keys	Lost or stolen
What you know and what you have	ATM + PIN	Shared + PIN is weak (written on back, easy to guess or forget)
What you are (something unique about the user)	Fingerprint	Not possible to share
	Face recognition	Reputation unlikely
	Voice print	Forging difficult
	Iris scan	Cannot be lost or stolen



Fig. 8: User using CRYPTOCARD (disconnected token) during authentication

storing the keys.

Some examples of 'what you have' are USB tokens [X.509 (PKI)], magnetic stripe cards, Dallas iButton, CASQUE, UniOTP, virtual tokens, audio port tokens, smartcards, ATM cards (swipe/PIN), mobile phones, SMSes, smartphone push, RFID, one-time pads and e-tokens (dongle with signa-

ture and RSA SecurID).

What you are

This type of authentication technique does not require a user to memorise a password because the password in this technique may be a finger, face, voice or eyes. However, these biometric properties alter over time. Applications used in this type of biometric authentication scan the face or finger first and then convert into text code or signature. This text code or signature is required to validate from an authorised system or server.

In the next issue, we will see how various websites have implemented two-factor authentication.

To be continued next month



Fig. 9: Two-factor authentication

connect with the system) have a digital display (SecurID token in Fig. 7) to show you the unique key every time you need to authenticate with the website. This factor is very prone to attacks. Keys can easily be stolen; if lost, a thief can use such keys to open the lock. So, it has become mandatory to use additional authentication or a secure way of